MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1CCRBCA100– Digital Fundamentals

Model Question Paper

Duration: 2hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions. Answer All Questions

Each question carries 2 marks

1. Explain the overflow condition in binary addition with a suitable example. [U][1]
2. Define Multiplexer. [U][3]
3. Draw the truth table for a three variable minterm representation F(x,y,z) = ∑(0,3,6). [U][1]
4. Distinguish the positive and negative edges of a clock pulse. [An][4]
5. Convert the binary number 11012 to its decimal equivalent. [A][1]
6. Write short notes on Maxterm and Minterm. [U][2]
7. How can a NOR gate be used to implement the AND function? [A][2]
8. Write short note on Combinational Logic Circuits and Sequential Logic Circuits. [U][4]
9. Draw the truth table of a JK flip-flop. [U][4]
10. What are the applications of an encoder in digital systems? [U][3]

(10*2=20 Marks)

Part B

Short Answer Questions (5 out of 7 Questions)

Each question carries 6 marks

11. Convert the decimal number 483 to its equivalent BCD using the 8421 code [A][1]
12. Draw the logic diagram using only NOR gates F=xy'z +x'yz'. [A][2]
13. Draw the circuit diagram of 1×8 De-multiplexer. [A][3]
14. What are the applications of shift registers? [U][4]
15. Convert the octal number 674 to hexadecimal. [A][1]
16. Convert to SOP (A' + B' + C) (B + C' + D')(A'+ B + C + D'). [A][2]
17. Describe T flip-flop. What is its use? [U][4]

(5*6=30 Marks)

Part C

Essay Questions (2 out of 3 Questions)

Each question carries 10 marks

18. Simplify using K-map, the expression F(A,B,C,D)= ∑m(0,2,7,8,14,) with Don't care terms D= ∑m(1,6,9,13). Draw the corresponding logic diagram. [A][2]
19. Design R-S flip-flop and JK flip-flop using NOR gates only. [U][4]

20. Explain the working of a digital encoder. Describe their applications and advantages.

[U][3]

(2*10=20 Marks)